Grade 11 – Term 1
C5, C6 and C7 Exam Solutions – Version 2

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Learning objective

Use continuous probability models and sample data to describe typical values, variability, and interval probabilities, and interpret differences between sample estimates and theoretical values.

Criteria assessment

This exam assesses C5, C6 and C7 separately, with 6 points per criterion (18 points in total). Each task shows its point allocation. Partial credit may be awarded when the correct procedure is clearly shown but arithmetic errors are made. Answers without the required procedure will not receive credit. Failure to write your name on the task sheet or submitting it after the deadline will result in a one-point deduction from each criterion. For this exam, a minimum of 4 out of 6 points is required to pass each criterion. Use unrounded intermediate values; round final statistics to three decimal places and percentage errors to two decimal places where necessary.

Evaluated criteria

C5 Calculates and interprets the mean, variance and standard deviation of continuous distributions using standard formulas and functions.

C6 Estimates interval probabilities using relative frequencies from generated samples. and compares them with theoretical probabilities.

C7 Calculates sample means, variances, and standard deviations for different sample sizes and data representations.

1 Two custom-print studios [C5-2]

Two studios print custom posters, with each order taking between 3 and 15 hours to complete. Both studios handle a wide variety of orders, but their production processes differ. Studio I specializes in large, detailed posters that require extensive customization. As a result, longer completion times are more common, with orders becoming progressively more likely to require additional production time. Studio D uses a more efficient printing process designed for quick orders. Shorter completion times are more common, and orders become progressively less likely to require additional production time. Let I and D represent the respective completion times in hours.

Questions/Tasks



C5 Identify each distribution from the clues and calculate its mean, variance, and standard deviation using the corresponding formulas. **[2]**

C5.1(a) – Linear models **[2]**

The intended models are a linearly increasing distribution for studio I and a linearly decreasing distribution for studio D. The contextual trends alone do not specify linear densities uniquely; the following results apply under these intended linear models. Thus $I \sim \text{Lin}_{\text{inc}}(3, 15)$ and $D \sim \text{Lin}_{\text{dec}}(3, 15)$, with $a = 3$, $b = 15$.

$$E(I) = \frac{a + 2b}{3} = \frac{3 + 2(15)}{3} = 11.000 \text{ h}, \qquad E(D) = \frac{2a + b}{3} = \frac{2(3) + 15}{3} = 7.000 \text{ h}.$$

$$Var(I) = Var(D) = \frac{(b - a)^2}{18} = \frac{12^2}{18} = 8.000 \text{ h}^2, \qquad \sigma_I = \sigma_D = \sqrt{8} \approx 2.828 \text{ h}.$$

2 Bookings and daily takings [C5-2]

A camera repair shop's daily revenue varies due to many small, independent factors, including customer demand and repair costs. Most days, revenue is close to \$3200, while higher and lower amounts become increasingly uncommon at similar rates. The standard deviation is \$200. Let R be daily revenue in dollars. The shop receives online bookings independently at an average rate of 10 per hour. Bookings arrive at a steady average rate, and the time since the last booking does not affect when the next arrives. Let T be the time between bookings in minutes.

Questions/Tasks



C5 Identify the distribution for T from the clues and use its formulas to find the mean, variance, and standard deviation in minutes. **[1]**

C5 Identify the revenue distribution from the clues. Use its formulas to find the mean, variance, and standard deviation in dollars. Interpret the mean. **[1]**

C5.2(a) – Exponential waiting time [1]

Independent arrivals at a steady average rate and memoryless waiting identify an exponential distribution. The rate is $\lambda = 10/60 = 1/6$ per minute, so $T \sim \text{Exp}(1/6)$.

$$E(T) = \frac{1}{\lambda} = 6.000 \text{ min}, \qquad Var(T) = \frac{1}{\lambda^2} = 36.000 \text{ min}^2, \qquad \sigma_T = \frac{1}{\lambda} = 6.000 \text{ min}.$$

C5.2(b) – Normal revenue and units [1]

The symmetric pattern and many small, independent influences suggest a normal model. Its mean is \$3200 and the stated standard deviation is $\sigma_R = 200$ dollars. Thus $R \sim N(3200, 40000)$ (mean, variance).

$$E(R) = \mu_R = 3200.000 \text{ dollars}, \qquad Var(R) = \sigma_R^2 = 200^2 = 40000.000 \text{ dollars}^2,$$

$$\sigma_R = \sqrt{Var(R)} = \sqrt{40000} = 200.000 \text{ dollars}.$$

The long-run average daily revenue is \$3200; individual days may differ.

3 Two parcel-labeling stations [C5-2]

A delivery company uses two parcel-labeling stations. At station A, an automated machine labels parcels in 5 to 19 minutes. Due to differences in parcel sizes and handling requirements, completion times vary unpredictably, with no particular duration being more common than another. Let T_A be its labeling time in minutes.

At station B, staff label special parcels, also taking between 5 and 19 minutes. Most parcels are completed in around 10 minutes, while shorter and longer processing times become progressively less common. Very quick jobs are especially rare, and longer jobs occasionally require additional handling. Let T_B be its labeling time in minutes.

Questions/Tasks

C5 Identify the distribution for T_A from the clues and calculate its mean, variance, and standard deviation using the corresponding formulas. **[1]**

C5 Identify the distribution for T_B from the clues and calculate its mean, variance, and standard deviation using the corresponding formulas. **[1]**

C5.3(a) – Uniform model [1]

No particular duration being more common identifies the uniform model $T_A \sim U(5, 19)$. With $a = 5$, $b = 19$, apply $\mu = (a + b)/2$, $\sigma^2 = (b - a)^2/12$, and $\sigma = \sqrt{\sigma^2}$:

$$E(T_A) = \frac{5 + 19}{2} = 12.000 \text{ min}, \qquad Var(T_A) = \frac{(19 - 5)^2}{12} = \frac{49}{3} \approx 16.333 \text{ min}^2, \qquad \sigma_{T_A} = \sqrt{49/3} \approx 4.041 \text{ min}.$$

C5.3(b) – Triangular model **[1]**

The intended model is triangular, with bounds 5 and 19 and mode 10 minutes. The contextual description alone does not uniquely specify a triangular density or an exact mode; using this intended model gives $T_B \sim \text{Tri}(5, 19, 10)$ (minimum, maximum, mode). Apply $\mu = (a+b+c)/3$ and $\sigma^2 = (a^2+b^2+c^2-ab-ac-bc)/18$:

$$E(T_B) = \frac{5+19+10}{3} = \frac{34}{3} \approx 11.333 \text{ min},$$

$$Var(T_B) = \frac{5^2+19^2+10^2-5(19)-5(10)-19(10)}{18} = \frac{486-335}{18} = \frac{151}{18} \approx 8.389 \text{ min}^2, \qquad \sigma_{T_B} = \sqrt{151/18} \approx 2.896 \text{ min}.$$

4 File upload times, grouped data [C6-2]

A sample of file upload times in seconds is summarized below. Classes are left-inclusive and right-exclusive.

Class	$[2,8)$	$[8,14)$	$[14,20)$	$[20,26)$
f_i	5	9	7	3

The theoretical probability is $P(8 \le X < 20) = 0.75$.

Questions/Tasks

C6 Find the empirical probability of the stated interval from the given sample. **[1]**

C6 Calculate the absolute and percentage errors relative to the given theoretical probability. **[1]**

C6.1(a) – Relative frequency **[1]**

The event contains the two middle classes: $n = 5+9+7+3 = 24$ and $k = 9+7 = 16$. Because the event is a union of complete classes, no individual upload times need to be estimated.

$$\widehat{P}(8 \le X < 20) = \frac{16}{24} = \frac{2}{3} \approx 0.667.$$

C6.1(b) – Errors **[1]**

Using the exact relative frequency,

$$E = \left|\frac{2}{3} - 0.75\right| = \frac{1}{12} \approx 0.083, \qquad \text{Percentage error} = \frac{\frac{1}{12}}{0.75} \times 100\% \approx 11.11\%.$$

5 Office electricity use, raw data [C6-1]

A sample of daily office electricity use over 20 working days (tens of kilowatt-hours) is

1.3	1.7	2.0	2.2	2.6	2.9	3.3	3.5	3.8	4.2
1.6	1.9	2.3	2.7	3.1	3.4	3.6	3.9	4.4	4.6

Let X be daily electricity use in tens of kilowatt-hours. The interval 20 to 35 kWh, including both limits, is $2 \le X \le 3.5$ in these units. The theoretical probability is $P(2 \le X \le 3.5) = 0.60$.

Questions/Tasks

C6 Find the empirical probability of the stated interval from the given sample. **[0.5]**

C6 Calculate the absolute and percentage errors relative to the given theoretical probability. **[0.5]**

C6.2(a) – Relative frequency [0.5]

There are $n = 20$ observations. The included observations are $2.0, 2.2, 2.6, 2.9, 3.3, 3.5, 2.3, 2.7, 3.1, 3.4$, so $k = 10$.

$$\widehat{P}(2 \le X \le 3.5) = \frac{10}{20} = \frac{1}{2} \approx 0.500.$$

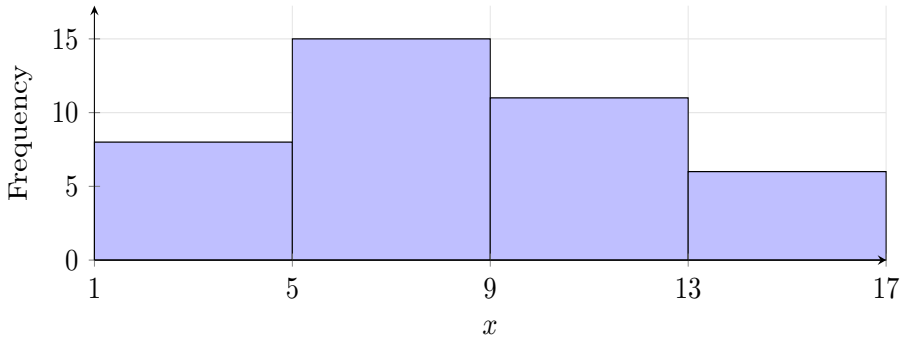
C6.2(b) – Errors [0.5]

Using the exact relative frequency,

$$E = \left|\frac{1}{2} - 0.60\right| = \frac{1}{10} \approx 0.100, \qquad \text{Percentage error} = \frac{\frac{1}{10}}{0.60} \times 100\% \approx 16.67\%.$$

6 Museum entrance queues, histogram data [C6-2]

The histogram is the only record of a sample of museum entrance queue times. Bar heights are frequencies; times are in minutes, with left-inclusive, right-exclusive classes.



The theoretical probability is $P(5 \le X < 13) = 0.70$.

Questions/Tasks

C6 Find the empirical probability of the stated interval from the given sample. **[1]**

C6 Calculate the absolute and percentage errors relative to the given theoretical probability. **[1]**

C6.3(a) – Relative frequency **[1]**

The classes $[1,5),[5,9),[9,13),[13,17)$ have frequencies $8,15,11,6$. Thus $n = 8+15+11+6 = 40$ and $k = 15+11 = 26$.

$$\widehat{P}(5 \le X < 13) = \frac{26}{40} = \frac{13}{20} \approx 0.650.$$

C6.3(b) – Errors **[1]**

Using the exact relative frequency,

$$E = \left|\frac{13}{20} - 0.70\right| = \frac{1}{20} \approx 0.050, \qquad \text{Percentage error} = \frac{\frac{1}{20}}{0.70} \times 100\% \approx 7.14\%.$$

7 Equipment rental times, frequency data [C6-1]

Recorded equipment rental times are summarized below.

x_i (min)	4	8	12	16	20	24
f_i	8	16	28	22	14	12

The theoretical probability is $P(8 \le X < 20) = 0.80$.

Questions/Tasks

C6 Find the empirical probability of the stated interval from the given sample. **[0.5]**

C6 Calculate the absolute and percentage errors relative to the given theoretical probability. **[0.5]**

C6.4(a) – Relative frequency **[0.5]**

The total is $n = 8+16+28+22+14+12 = 100$. The value 20 is excluded, so $k = 16+28+22 = 66$.

$$\widehat{P}(8 \le X < 20) = \frac{66}{100} = \frac{33}{50} \approx 0.660.$$

C6.4(b) – Errors **[0.5]**

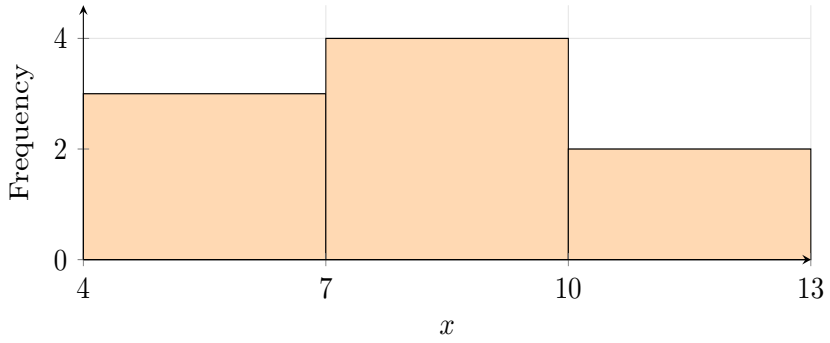
Using the exact relative frequency,

$$E = \left|\frac{33}{50} - 0.80\right| = \frac{7}{50} \approx 0.140, \qquad \text{Percentage error} = \frac{\frac{7}{50}}{0.80} \times 100\% \approx 17.50\%.$$

Use the sample-variance denominator $n-1$ throughout these four problems. For each statistic, calculate absolute error $E = |\text{sample value} - \text{theoretical value}|$ and percentage error $100E/|\text{theoretical value}|$. Use unrounded values in all error calculations. For grouped data and histograms, use class midpoints to estimate the statistics.

8 Luggage masses, histogram data [C7-2]

A histogram gives a sample of luggage masses in kilograms. Bar heights are frequencies; classes are left-inclusive and right-exclusive. The model has $\mu = 9$ kg and $\sigma^2 = 7$ kg^2.



Questions/Tasks

C7 Using the grouped data, estimate the sample mean, sample variance, and sample standard deviation. **[1]**

C7 Calculate the absolute and percentage errors of the mean, variance, and standard deviation relative to the theoretical values. **[1]**

C7.1(a) – Histogram statistics [1]

The class midpoints are $5.5, 8.5, 11.5$ kg. Use these as the representative observations.

$$n = 3 + 4 + 2 = 9, \qquad \bar{x} \approx \frac{3(\frac{11}{2}) + 4(\frac{17}{2}) + 2(\frac{23}{2})}{9} = \frac{49}{6} \approx 8.167.$$

The weighted squared-deviation sum, using the unrounded mean, is

$$3\left(\frac{11}{2} - \frac{49}{6}\right)^2 + 4\left(\frac{17}{2} - \frac{49}{6}\right)^2 + 2\left(\frac{23}{2} - \frac{49}{6}\right)^2 = 44.$$

$$s^2 \approx \frac{44}{9-1} = \frac{11}{2} \approx 5.500, \qquad s \approx \sqrt{\frac{11}{2}} \approx 2.345.$$

The mean and standard deviation are in kg; the variance is in squared kg.

C7.1(b) – Absolute and percentage errors [1]

The theoretical standard deviation is $\sigma = \sqrt{7}$. Apply $E = |\text{sample value} - \text{theoretical value}|$ and $100E/|\text{theoretical value}|$ to the unrounded results:

Statistic	Absolute error	Percentage error
Mean	0.833	9.26%
Variance	1.500	21.43%
Standarddeviation	0.301	11.36%

Absolute errors have the units of the corresponding statistic; percentage errors are unitless. These errors use midpoint estimates because individual observations are unknown.

9 Chamber temperatures, raw data [C7-1]

Five chamber temperature readings in degrees Celsius are $14, 17, 18, 19, 22$. The theoretical model has $\mu = 19$ degrees Celsius and $\sigma^2 = 6$ square degrees Celsius.

Questions/Tasks

C7 Calculate the sample mean, sample variance, and sample standard deviation from the given data. **[0.5]**

C7 Calculate the absolute and percentage errors of the mean, variance, and standard deviation relative to the theoretical values. **[0.5]**

C7.2(a) – Raw-data statistics [0.5]

$$n = 1+1+1+1+1 = 5, \qquad \bar{x} = \frac{1(14)+1(17)+1(18)+1(19)+1(22)}{5} = 18 \approx 18.000.$$

x_i	14	17	18	19	22
$x_i - \bar{x}$	-4	-1	0	1	4
$(x_i - \bar{x})^2$	16	1	0	1	16

The squared deviations sum to $16 + 1 + 0 + 1 + 16 = 34$.

$$s^2 = \frac{34}{5-1} = \frac{17}{2} \approx 8.500, \qquad s = \sqrt{\frac{17}{2}} \approx 2.915.$$

The mean and standard deviation are in degrees Celsius; the variance is in squared degrees Celsius.

C7.2(b) – Absolute and percentage errors [0.5]

The theoretical standard deviation is $\sigma = \sqrt{6}$. Apply $E = |\text{sample value} - \text{theoretical value}|$ and $100E/|\text{theoretical value}|$ to the unrounded results:

Statistic	Absolute error	Percentage error
Mean	1.000	5.26%
Variance	2.500	41.67%
Standarddeviation	0.466	19.02%

Absolute errors have the units of the corresponding statistic; percentage errors are unitless.

10 Parcels dispatched, frequency data [C7-1]

The numbers of parcels dispatched in nine short collection periods are summarized below. The theoretical model has $\mu = 7$ parcels and $\sigma^2 = 9$ parcels^2.

x_i	3	7	11
f_i	2	4	3

Questions/Tasks

C7 Calculate the sample mean, sample variance, and sample standard deviation from the given data. **[0.5]**

C7 Calculate the absolute and percentage errors of the mean, variance, and standard deviation relative to the theoretical values. **[0.5]**

C7.3(a) – Weighted statistics [0.5]

$$n = 2 + 4 + 3 = 9, \qquad \bar{x} = \frac{2(3) + 4(7) + 3(11)}{9} = \frac{67}{9} \approx 7.444.$$

The weighted squared-deviation sum, using the unrounded mean, is

$$2\left(3 - \frac{67}{9}\right)^2 + 4\left(7 - \frac{67}{9}\right)^2 + 3\left(11 - \frac{67}{9}\right)^2 = \frac{704}{9}.$$

$$s^2 = \frac{\frac{704}{9}}{9 - 1} = \frac{88}{9} \approx 9.778, \qquad s = \sqrt{\frac{88}{9}} \approx 3.127.$$

The mean and standard deviation are in parcels; the variance is in squared parcels.

C7.3(b) – Absolute and percentage errors [0.5]

The theoretical standard deviation is $\sigma = \sqrt{9}$. Apply $E = |\text{sample value} - \text{theoretical value}|$ and $100E/|\text{theoretical value}|$ to the unrounded results:

Statistic	Absolute error	Percentage error
Mean	0.444	6.35%
Variance	0.778	8.64%
Standarddeviation	0.127	4.23%

Absolute errors have the units of the corresponding statistic; percentage errors are unitless.

11 Walking times, grouped data [C7-2]

Walking times in minutes are grouped using left-inclusive, right-exclusive classes. The theoretical mean is 23 minutes and variance is 30 minutes^2.

Time (min)	$[12, 18)$	$[18, 24)$	$[24, 30)$
f_i	2	3	4

Questions/Tasks

C7 Using the grouped data, estimate the sample mean, sample variance, and sample standard deviation. **[1]**

C7 Calculate the absolute and percentage errors of the mean, variance, and standard deviation relative to the theoretical values. **[1]**

C7.4(a) – Midpoint estimates **[1]**

The class midpoints are $15, 21, 27$ min. Use these as the representative observations.

$$n = 2 + 3 + 4 = 9, \qquad \bar{x} \approx \frac{2(15) + 3(21) + 4(27)}{9} = \frac{67}{3} \approx 22.333.$$

The weighted squared-deviation sum, using the unrounded mean, is

$$2\left(15 - \frac{67}{3}\right)^2 + 3\left(21 - \frac{67}{3}\right)^2 + 4\left(27 - \frac{67}{3}\right)^2 = 200.$$

$$s^2 \approx \frac{200}{9 - 1} = 25 \approx 25.000, \qquad s \approx \sqrt{25} \approx 5.000.$$

The mean and standard deviation are in min; the variance is in squared min.

C7.4(b) – Absolute and percentage errors **[1]**

The theoretical standard deviation is $\sigma = \sqrt{30}$. Apply $E = |\text{sample value} - \text{theoretical value}|$ and $100E/|\text{theoretical value}|$ to the unrounded results:

Statistic	Absolute error	Percentage error
$Mean$	0.667	2.90%
$Variance$	5.000	16.67%
$Standarddeviation$	0.477	8.71%

Absolute errors have the units of the corresponding statistic; percentage errors are unitless. These errors use midpoint estimates because individual observations are unknown.



12 Summarized Solutions

Notation and rounding. $\mu = E(X)$, $\sigma^2 = \mathrm{Var}(X)$, σ = theoretical SD; $\bar{x}, s^2, s$ = sample mean, variance, SD. Decimal columns show three places (rounded where necessary); percentage errors show two. Model parameters, counts and fractions are exact. All errors use unrounded values.

C5

Distribution models and theoretical statistics

Question	Variable/model	$E(X)$	$\mathrm{Var}(X)$	σ	Units
1, I	$I \sim \mathrm{Lin}_{\mathrm{inc}}(3, 15)$	11.000	8.000	2.828	h; h^2
1, D	$D \sim \mathrm{Lin}_{\mathrm{dec}}(3, 15)$	7.000	8.000	2.828	h; h^2
2, T	$T \sim \mathrm{Exp}(1/6)$	6.000	36.000	6.000	min; min^2
2, R	$R \sim N(3200, 40000)$	3200.000	40000.000	200.000	dollars; $\mathrm{dollars}^2$
3, A	$T_A \sim U(5, 19)$	12.000	16.333	4.041	min; min^2
3, B	$T_B \sim \mathrm{Tri}(5, 19, 10)$	11.333	8.389	2.896	min; min^2

Mean and variance: substitute before calculating

Question	$E(X)$: substituted expression	$\mathrm{Var}(X)$: substituted expression
1, I	$\frac{3 + 2(15)}{3}$	$\frac{(15 - 3)^2}{18}$
1, D	$\frac{2(3) + 15}{3}$	$\frac{(15 - 3)^2}{18}$
2, T	$\frac{1}{1/6}$	$\frac{1}{(1/6)^2}$
2, R	$\mu_R = 3200$	200^2
3, A	$\frac{5 + 19}{2}$	$\frac{(19 - 5)^2}{12}$
3, B	$\frac{5 + 19 + 10}{3}$	$\frac{5^2 + 19^2 + 10^2 - 5(19) - 5(10) - 19(10)}{18}$

Units: first for $E(X), \sigma$; second for $\mathrm{Var}(X)$. Conventions: $N(\mu, \sigma^2)$; $\mathrm{Tri}(a, b, c)$ = (minimum, maximum, mode); exponential rate $\lambda = 1/6$ min^{-1}.

Model notes. Question 1: intended increasing/decreasing linear densities on $[3, 15]$; the contextual trends do not uniquely determine linear models. Question 3: intended triangular density with mode 10 min; the description alone specifies neither a unique triangular density nor an exact mode. Question 2: \$3200 is the long-run average daily revenue; individual days may differ.

C6

Empirical probabilities and errors

P_0 = theoretical probability, $E = |\widehat{P} - P_0|$, $E\% = 100E/|P_0|$. For each question: $n, k, \widehat{P}$ and errors $E, E\%$.

Question	Event	n	k	$\widehat{P} = k/n$	P_0	E	$E\%$
4	$8 \le X < 20$ s	24	16	$\frac{16}{24} \approx 0.667$	0.750	0.083	11.11%
5	$2 \le X \le 3.5$	20	10	$\frac{10}{20} = 0.500$	0.600	0.100	16.67%
6	$5 \le X < 13$ min	40	26	$\frac{26}{40} = 0.650$	0.700	0.050	7.14%
7	$8 \le X < 20$ min	100	66	$\frac{66}{100} = 0.660$	0.800	0.140	17.50%

Question 5: X is in tens of kWh; $2 \le X \le 3.5$ means $20 \le$ use ≤ 35 kWh. Questions 4 and 6: complete classes give exact counts; no midpoint estimates. Probabilities and their errors are unitless.

C7

Sample statistics and comparison with theory

$s^2 = \dfrac{\sum_i f_i (x_i - \bar{x})^2}{n-1}$, with $f_i = 1$ for raw data; $\quad E = |\text{sample value} - \text{theoretical value}|, \quad E\% = 100E/|\text{theoretical value}|$.

Sample statistics

Question	n	$\bar{x}$	s^2	s	Units
8	9	8.167	5.500	2.345	kg; kg^2
9	5	18.000	8.500	2.915	°C; $(°\text{C})^2$
10	9	7.444	9.778	3.127	parcels; parcels^2
11	9	22.333	25.000	5.000	min; min^2

Units: first for $\bar{x}, s$; second for s^2. *Midpoint estimates:* Question 8 uses $(5.5, 8.5, 11.5)$ kg; Question 11 uses $(15, 21, 27)$ min. Their statistics and errors in both tables are based on these midpoint estimates because individual observations are unknown.

Errors relative to theory

Question	Statistic	Sample value	Theoretical value	E	$E\%$
8	$\bar{x}$ vs. μ (kg)	8.167	9.000	0.833	9.26%
	s^2 vs. σ^2 (kg^2)	5.500	7.000	1.500	21.43%
	s vs. σ (kg)	2.345	$\sqrt{7} \approx 2.646$	0.301	11.36%
9	$\bar{x}$ vs. μ (°C)	18.000	19.000	1.000	5.26%
	s^2 vs. σ^2 ($(°\text{C})^2$)	8.500	6.000	2.500	41.67%
	s vs. σ (°C)	2.915	$\sqrt{6} \approx 2.449$	0.466	19.02%
10	$\bar{x}$ vs. μ (parcels)	7.444	7.000	0.444	6.35%
	s^2 vs. σ^2 (parcels^2)	9.778	9.000	0.778	8.64%
	s vs. σ (parcels)	3.127	3.000	0.127	4.23%
11	$\bar{x}$ vs. μ (min)	22.333	23.000	0.667	2.90%
	s^2 vs. σ^2 (min^2)	25.000	30.000	5.000	16.67%
	s vs. σ (min)	5.000	$\sqrt{30} \approx 5.477$	0.477	8.71%

Sample values, theoretical values and absolute errors E have the units shown in the Statistic column; percentage errors are unitless.